



CROSS-EMBODIMENT LOCOMOTION TRANSFER FROM EGOCENTRIC
VISUAL WORLD MODEL
WITHOUT PRIOR KNOWLEDGE OF THE MORPHOLOGY

MISS DISTHORN SUTTAWET

A THESIS SUBMITTED IN PARTIAL FULFILLMENT
OF THE REQUIREMENTS FOR
THE DEGREE OF BACHELOR OF ENGINEERING
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Abstract

Reinforcement-learning controllers for legged locomotion are tightly coupled to the body they were trained on, and every existing route around this hands the model privileged information about the new body – a CAD/URDF description, a kinematic tree, a fitted per-robot adapter, or demonstrations of the target. This research tests whether a behaviour recorded on one robot can instead drive a *structurally different* robot from egocentric video alone, through a **shared body-motion coordinate** (Froude-scaled forward, lateral, and yaw velocity) with no kinematic model, no URDF, and no demonstrations of the target. The two bodies – an 18-DOF stick insect (*Medauroidea extradentata*) in CoppeliaSim and a 12-DOF Unitree B1 quadruped in MuJoCo – share no joint dimension or correspondence; a frozen V-JEPA2 encoder feeds an Inverse Transition Model, a Forward Transition Model, and a Motion Decoder, adapting the latent-action world-model architecture from manipulation to legged locomotion. Results to date show the coordinate transfers can be *correspondence-free*, with no frame pairing or retargeting between the two bodies. Because locomotion is periodic, a third-person view lets the robot’s own pose stand in for the command, so the world model predicts by completing the gait cycle rather than using the learned latent action, while an egocentric view restores measurable action-conditioning. The established limitation is that this conditioning stays coarse: the model uses the action at a single step but does not yet rank fine differences between similar actions or roll out reliably over long horizons.

Keywords: cross-embodiment locomotion / egocentric vision / world model / body-motion coordinate / V-JEPA2

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LIST OF SYMBOLS

o_t	RGB observation (frame) at time t
e_t	Visual embedding of frame t (frozen encoder output)
z_t	Latent action inferred from the transition $e_t \rightarrow e_{t+1}$
a_t	Native joint command (18-D joint-position target)
$\hat{e}_{t+1}$	Predicted next embedding (Forward Transition Model output)
$\hat{a}_t$	Predicted joint command (Motion Decoder output)
λ	Loss-weighting coefficient
δ	Cliff's delta (effect-size measure)

LIST OF ABBREVIATIONS

RL	Reinforcement Learning
WM	World Model
CM-VLAWM	Cross-Morphology Visual Latent Action World Model (this work)
LAC-WM	Latent Action-Conditioned World Model
ITM	Inverse Transition Model
FTM	Forward Transition Model
MD	Motion Decoder
IK	Inverse Kinematics
CPG	Central Pattern Generator
JEPA	Joint-Embedding Predictive Architecture
AEP/PEP	Anterior / Posterior Extreme Position

CHAPTER 1 INTRODUCTION

1.1 Background and Significance

Legged robots are typically controlled by policies learned through reinforcement learning (RL). A persistent and expensive limitation of this approach is **morphology specificity**: a policy learned for one body does not generalize to another. If a robot’s legs are lengthened or shortened, its mass distribution changes, or its skeleton is a different topology altogether, the previously trained policy fails and a new one must be learned from scratch, a process that can take hours to days of training per body. In contrast, biological organisms share locomotion principles across vastly different body plans, suggesting that a shared, body-independent representation of *movement* is possible.

The precise analogy for what this thesis attempts already has a name in psychology: **vicarious learning** (Bandura, 1977), the acquisition of a behaviour by observing another agent perform it, with no direct instruction, no first-hand demonstration of the observer’s own body doing the task, and no requirement that the observed model resemble the observer. That is exactly the transfer under test here: the quadruped’s controller is informed by video of a stick insect walking, a body it does not resemble and has never itself performed that behaviour on, with no direct instruction and no kinematic account of either body supplied to bridge them. The behaviour crosses through what is observed, not through anything told to the model about the bodies involved.

Every existing route around this hands the model privileged information about the new body. The strategies differ mainly in where that information comes from: a CAD or URDF description read directly from the robot’s design files (QWM); a kinematic tree specifying which joint connects to which (graph- and transformer-based universal controllers); an observation encoder and action decoder fitted per robot from that robot’s own proprioception and foot force (L3P); a task space such as end-effector pose that already means the same thing on every body in the corpus (LAC-WM); or demonstrations of the same task paired across both bodies (Demo-JEPA). Each is reasonable, and each assumes access to something about the target body that the motivating scenario – an animal, a damaged robot, or hardware acquired without published kinematics – does not provide.

One result shows the requirement can be dropped entirely, and it is the starting point for this thesis. Hu, Chen, and Lipson (2025) learn a task-agnostic visual self-model for a legged robot from a single egocentric camera and random motor babbling, with no prior knowledge of the robot’s morphology, kinematics, or task, and use it to plan locomotion and even to detect and recover from physical damage. Their claim is that the body description was never necessary. But their self-model is fitted to *one robot at a time*: each body is babbled and modelled separately, there is no action representation shared between bodies, and nothing is ever transferred from one body to another. **Extending that**

premise across embodiments is what this thesis attempts, and it is the step nobody in this literature has taken.

Why this cannot be answered on bodies that share one skeleton. An earlier phase of this project asked whether the same idea transfers to legged locomotion using three instances of one stick-insect body differing only in leg length. That setting is useful as a controlled first step (Section 1.5, Stage 1) but it cannot, on its own, establish that *vision* is doing something *proprioception* could not: three leg lengths share an identical 18-dimensional joint space, so a proprioceptive model could in principle be shared across them too, and vision’s advantage there is convenience, not necessity. Answering the harder question, whether a vision-only latent action can unify locomotion across bodies where *proprioception has no shared coordinate to begin with*, requires a second body whose action space is genuinely **disjoint** from the first. This thesis therefore studies two robots at once: a simulated six-legged stick insect (*Medauroidea extradentata*, 18 actuated joints, in CoppeliaSim) and a Unitree B1 quadruped (12 actuated joints, in MuJoCo). Their joint spaces share no dimension and no correspondence; a single egocentric camera, by contrast, describes both bodies in the same $256 \times 256 \times 3$ pixel array regardless of what is standing in front of it.

The object this thesis learns is not, in the end, a single opaque latent action but a **shared body-motion coordinate**: dimensionless forward, lateral, and yaw velocity, the same three physical quantities on both robots, inferred from egocentric video alone, with no morphology label, no kinematic model or URDF for either body, and no manually defined correspondence between one robot’s joints and the other’s. Learning this coordinate is one problem; *using* it to drive an unseen or unfamiliar body is a second, harder problem, and a substantial part of this thesis is the diagnostic work that separates what the resulting world model can do (coarse, single-step action-conditioning) from what it cannot yet do reliably (fine-grained action ranking and multi-step rollout), and the measured mechanism behind that gap.

Pursuing that second problem exposed something the prior work never had to confront, and it is the scientific contribution of this thesis. Hu et al. adopt an egocentric camera for practical reasons – one onboard sensor, portability, ground texture supplying motion cues – and never compare it against a third-person view; no method surveyed in Chapter 2 asks whether the viewpoint matters at all. It does, and structurally. Locomotion is periodic, so an external view of a walking body shows a pose that is itself a function of gait phase, which makes the command largely readable from the body’s own posture. A world model trained on such video can therefore satisfy its prediction objective by continuing the gait cycle it can already see, without ever learning what the action does – and this thesis measures that it does exactly this. An egocentric view removes most of that redundancy and restores measurable action-conditioning. The viewpoint from which a locomotion world model is built is therefore not a free setup choice but a precondition for the model to use its action at all, which matters here because in cross-embodiment

transfer the action is precisely what has to cross.

The significance is twofold. Scientifically, it tests whether a body-independent notion of locomotion behaviour can be learned from visual observation alone, across bodies whose action spaces cannot be reconciled by any choice of coordinate without a kinematic model, and it produces, along the way, a diagnostic methodology for telling whether a video world model is *using* an inferred action at all rather than merely being able to decode it. Practically, if such a coordinate transfers cheaply, it would cut the cost of controlling a new robot by reusing locomotion behaviour already observed on a different body, rather than retraining from zero or requiring that body's engineering specification.

1.2 Objectives

1. To learn a shared, dimensionless body-motion coordinate (forward, lateral, and yaw velocity) from egocentric simulation video of two structurally disjoint legged robots (an 18-DOF hexapod and a 12-DOF quadruped), with no morphology label, no kinematic model or URDF for either body, and no frame-level correspondence assumed between them, by adapting the Latent Action World Model pipeline (frozen visual encoder $\rightarrow$ Inverse Transition Model $\rightarrow$ Forward Transition Model $\rightarrow$ Motion Decoder) from manipulation to locomotion.
2. To demonstrate that this coordinate transfers **correspondence-free** across the two embodiments – that a goal observed on one robot’s video can be used to select or drive behaviour on the other – and to show under which training objective this transfer succeeds and under which it silently fails (a reconstruction-only objective is shown to discard the action channel across embodiment families; a discriminative objective is required to recover it).
3. To characterize, through closed-loop diagnostic testing, exactly which capability of a video world model’s action-conditioning is and is not achieved: coarse single-step conditioning, fine-grained action ranking between similar candidates, and multi-step rollout, and to identify the measured mechanism (camera viewpoint and gait periodicity) responsible for the gap between them.

1.3 Expected Benefits

1. A demonstration that latent-action world models extend beyond manipulation to legged locomotion across bodies whose action spaces are genuinely disjoint, not merely differently parameterised instances of one topology.
2. Evidence on whether a shared, body-independent locomotion coordinate can be learned **without** privileged morphology information (no CAD/URDF/spec input) or a hand-defined cross-embodiment correspondence, using only egocentric video and each body’s own auto-logged motion.
3. A reusable diagnostic methodology for distinguishing whether a video world model’s inferred action is *legible* (decodable by some readout) from whether it is *used* (changes the model’s own prediction) – a distinction this thesis shows does not hold in general, and which prior displacement-based evidence for action-sensitivity does not by itself establish.
4. A stated and measured account of what a candidate-scoring deployment strategy requires of an “unseen” robot (a pre-existing policy able to produce the candidate behaviours), why that requirement is circular for the problem this thesis poses, and what is required instead (a policy trained to only actuate the robot, not to already know how to perform the target behaviours).

1.4 Scope of the Research

1. The study is conducted **entirely in simulation**: the hexapod in CoppeliaSim (Bullet physics engine) and the Unitree B1 quadruped in MuJoCo. No physical-robot or sim-to-real deployment is in scope.
2. **Two stages, two different questions.** Stage 1 instantiates one hexapod body as three leg-length variants sharing an identical 18-dimensional joint space (Section 3.5), and establishes that a learned latent action can beat a raw-joint-command baseline and transfer to a held-out leg length – a controlled result, but one that cannot by itself show vision succeeds where proprioception could not, because the three variants share a joint space proprioception could in principle share too. Stage 2 adds the Unitree B1 quadruped (12 actuated joints, MuJoCo), whose action space shares no dimension and no correspondence with the hexapod’s, and is the setting in which the central claim is tested: a shared body-motion coordinate, inferred from egocentric video alone, transfers where no proprioceptive coordinate could be shared at all without first being handed a kinematic model of both bodies.
3. The camera is **egocentric** (mounted on the body, first-person view), not the third-person / allocentric view used in the project’s earlier phase. This choice is itself a measured methodological finding (Section 3.2) rather than an arbitrary setup detail: an allocentric camera lets the robot’s own visible pose encode the command being executed (because locomotion is periodic), which lets a video world model satisfy its prediction objective by completing the observed gait cycle without ever reading the inferred action. An egocentric camera removes most of this pose-driven redundancy and is the setting in which the model is shown to measurably use the action at all.
4. The behaviours studied are drawn from a curated library of recorded conditions spanning forward, turning, and (on the hexapod) sideways locomotion at several speeds, not from an unconstrained action space; Section 3.6 states plainly what this library assumes about the target robot and why removing that assumption (via random motor babbling) is future work rather than a completed experiment.
5. The visual encoder (V-JEPA2) is used **frozen** throughout both stages; training it is out of scope.

Explicitly not claimed. To set expectations precisely, the following are outside the claims of this thesis: real-robot or animal transfer (the study is entirely in simulation); zero-shot cross-embodiment transfer of the forward model without any target-robot adaptation (Section 3.6.3 shows this fails, and reports cheap few-shot adaptation instead as the honest claim); reliable multi-step closed-loop control purely from the learned action (Section 3.7 shows action-conditioning holds coarsely and at one step, but not for fine-grained ranking or for rollouts beyond a few steps); a scaling law across many pretraining embodiments in the manner LAC-WM reports (only two embodiments exist in this study, which is a stated limitation, not a result); and the claim that a camera is the only information this pipeline needs about a new robot (the current deployment mechanism still draws on a

curated library of that robot’s own recorded behaviours, and Section 3.6.4 states why that is a real, unresolved limitation rather than a detail).

1.5 Research Procedure

The project follows a staged, milestone-gated procedure in which each stage’s result constrains the design of the next (this component-by-component validation strategy was set by the lab advisor):

1. **Stage 1 – cross-morphology (prerequisite).** Confirm the pipeline on three leg-length variants of one hexapod body: that identical commands produce measurably different behaviour across leg lengths, that the frozen visual encoder carries the behaviour signal needed downstream, and that a trained latent action beats a raw-joint-command baseline and transfers to a held-out leg length. This stage establishes that the pipeline works and that a latent action is better than the alternative; it does not establish that vision succeeds where proprioception could not, because all three bodies share one joint space (Section 1.1).
2. **Stage 2 – cross-embodiment.** Add the Unitree B1 quadruped, whose 12-dimensional action space shares no correspondence with the hexapod’s 18. Train a shared visual world-model trunk with per-embodiment decoder heads, and test whether a body-motion coordinate learned this way transfers correspondence-free between the two robots, and under which training objective (Section 3.4).
3. **Closed-loop diagnostic.** Build a candidate-scoring control loop – the world model ranks a library of recorded target-robot behaviours against a goal clip – as a test harness for the world model’s action-selection ability, while explicitly flagging why candidate scoring cannot itself be the final deployment mechanism for a genuinely unseen robot (Section 3.6.4).
4. **Root-cause diagnosis.** Where the closed loop’s behaviour does not match what the offline metrics predicted, isolate the mechanism through targeted ablations (camera viewpoint, gait periodicity, objective choice) rather than accepting the aggregate closed-loop number as the final word.
5. **Characterize the current limit and the path past it.** State precisely which action-selection capability the resulting world model has (coarse conditioning) and has not yet demonstrated (fine ranking, long-horizon rollout), and state the design (a trained policy rather than a scored candidate library) that removes the circularity in the deployment mechanism, as the concrete next step.

CHAPTER 2 LITERATURE REVIEW

This chapter builds toward a single question. Section 2.1 introduces only the machinery the rest of the thesis actually uses. Section 2.2 organises the cross-embodiment literature on one axis – *what each method must be told about a robot before it can control it* – because that is the axis this thesis’s claim lives on. Section 2.3 then states what crossing between two *structurally different* legged robots additionally requires beyond that axis: a coordinate that is not joint space. Section 2.4 covers the one result that removes the body-description requirement entirely, for a single robot, and is the closest precedent to this work. Section 2.5 is the pivot the rest of the thesis is built on: it asks whether a video world model conditioned on such a coordinate actually *uses* it, shows that the literature has not established this for locomotion, and states the two-part question – viewpoint, then precision – that Chapter 3 answers. Section 2.6 states the resulting gap and the claim of this thesis.

2.1 Theoretical Background

This section covers the machinery the argument depends on, and only that. A reader familiar with world models and self-supervised video representation learning can proceed to Section 2.2.

2.1.1 World models and learning in imagination

A *world model* is a learned model of environment dynamics: given the current state and an action, it predicts the next state. Once such a model exists, an agent can evaluate or train behaviour by “imagining” sequences of hypothetical actions inside it, rather than executing them. The reference implementation is DreamerV3 (Hafner et al., 2023), which encodes each observation into a compact latent state and learns the dynamics between latent states, so that the policy is trained against the model’s predictions rather than against the environment (Figure 2.1). A single agent with one hyperparameter setting mastered more than 150 tasks this way, which established learned latent dynamics as a general tool.

Two properties matter here. First, the representation the model learns is the object a downstream policy consumes, so any property that representation has – including independence from the body – is inherited by everything built on it. Second, and central to this thesis, a world model must answer “given this state and this action, what happens next?”, which makes its prediction objective an operational *test* of whether a proposed action representation carries meaningful information about the change it is supposed to explain. This thesis uses a world model less to imagine long rollouts than to use that test.

The limitation that motivates everything after this is that DreamerV3 and its relatives learn a separate model per domain and condition on the robot’s own motor command.

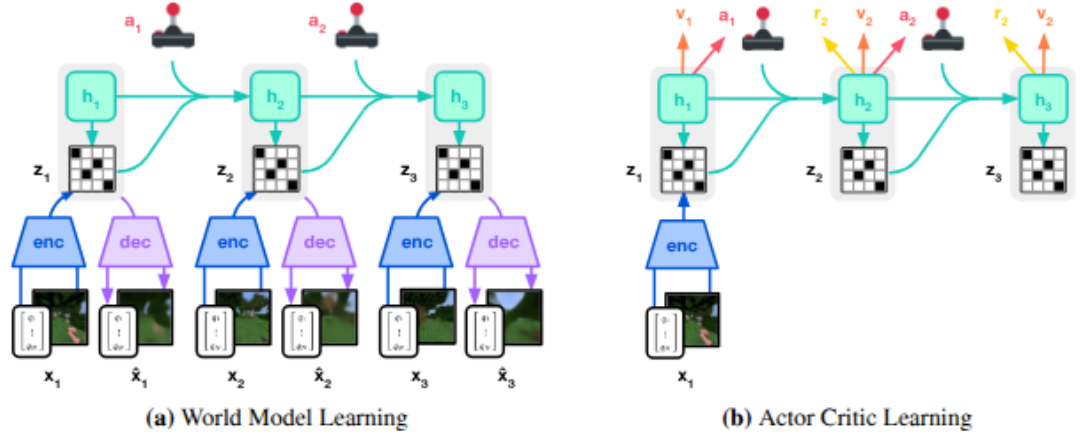


Figure 2.1 The DreamerV3 world model and actor-critic (Hafner et al., 2023). Observations are encoded into a compact latent state whose evolution a sequence model predicts from actions; the actor and critic then learn on trajectories *imagined* inside the model. Predictions are conditioned on the agent’s *native* action – the body-specific signal this thesis replaces.

Nothing is shared across bodies, and the native command is assumed to be the meaningful conditioning signal. When the same command means something different on a different body – or when two bodies have no command in common at all – that assumption fails.

2.1.2 Predicting representations rather than pixels

To apply a world model to images, one needs features. This work uses V-JEPA 2 (Assran et al., 2025), a video model trained on roughly one million hours of unlabeled video. Its objective is the distinguishing feature: instead of reconstructing masked pixels, a Joint-Embedding Predictive Architecture (JEPA) masks part of the input and predicts the **representation** of the masked part. Predicting in representation space frees the model from modelling unpredictable low-level detail and pushes it toward features that capture how things move and change (Figure 2.2).

Two aspects are directly relevant. The accompanying V-JEPA 2-AC variant uses the pretrained encoder as a **frozen per-frame image encoder** feeding a lightweight downstream predictor, which is the precedent for how the encoder is used here (Section 3.3). And V-JEPA 2 was trained on real-world video only, so its behaviour under a rendering domain gap is uncharacterised – which is why this project verifies empirically that the frozen features carry the required locomotion signal before relying on them.

2.1.3 Inverse and forward dynamics, and the latent action

Two complementary models recur throughout this literature and form this project’s architecture. A **forward dynamics model** maps (s_t, a_t) to $\hat{s}_{t+1}$: a world model’s transition predictor is exactly this. An **inverse dynamics model** maps (s_t, s_{t+1}) to $\hat{a}_t$, answering “what action caused this transition?” Neither is any single paper’s invention: inverse

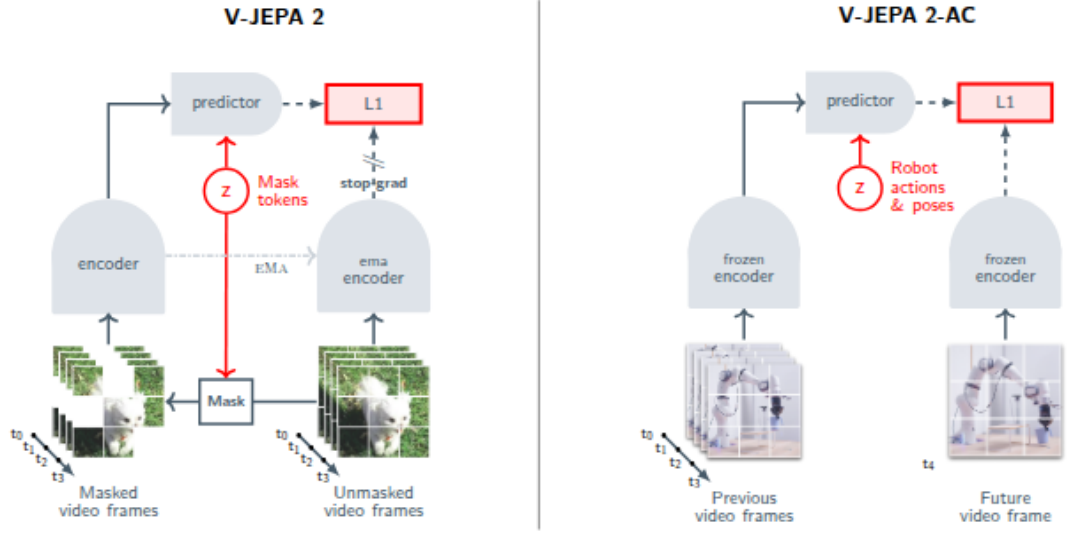


Figure 2.2 V-JEPA 2 multistage training (Assran et al., 2025). Left: self-supervised pretraining on internet-scale video with a mask-prediction objective in *representation* space. Right: the encoder is frozen and an action-conditioned predictor is learned on top. This project reuses only the frozen encoder as a per-frame visual front-end.

models were used by the Intrinsic Curiosity Module (Pathak et al., 2017) and by Video PreTraining (Baker et al., 2022), which labelled unannotated gameplay video with the actions that must have produced it.

The two combine into a *latent action*. If the quantity the inverse model recovers is not a native motor command but an inferred code z_t , and the forward model must predict the next state from the current state plus z_t , then the pair jointly learns an action representation: the inverse model proposes it and the forward model’s prediction error tests whether it suffices to explain the transition. Because z_t is defined by *observed change* rather than by any robot’s motor format, one latent space can in principle describe several bodies, and it can be learned from video with no action labels at all. Genie (Bruce et al., 2024) learned a discrete latent action from unlabeled internet video; LAPA (Ye et al., 2024) carried the idea toward real robot control; CLAM (Liang et al., 2025) introduced *continuous* latent actions, which suit the smooth, graded control locomotion requires; and Zhang et al. (2025) analyse what such models actually capture, which matters when interpreting whether a latent encodes behaviour or merely copies the next observation.

2.1.4 Latent action world models: the reference architecture

LAC-WM (Huang et al., 2026) combines the latent action with a world model to unify one action space across heterogeneous manipulation embodiments, and is the concrete architecture this project adapts (Figure 2.3). It has four parts: a **frozen visual encoder** producing e_t ; an **Inverse Transition Model (ITM)** mapping (e_t, e_{t+1}) to the latent action z_t ; a **Forward Transition Model (FTM)** predicting $\hat{e}_{t+1}$ from (e_t, z_t) ; and a **Motion Decoder (MD)** mapping z_t back to executable motor commands, which anchors the latent to real motion so it cannot collapse into a meaningless code. The inverse/forward/decoder

skeleton is the general building block of the preceding subsection rather than LAC-WM’s invention; what this project takes from LAC-WM specifically is its instantiation of that skeleton for cross-embodiment transfer.

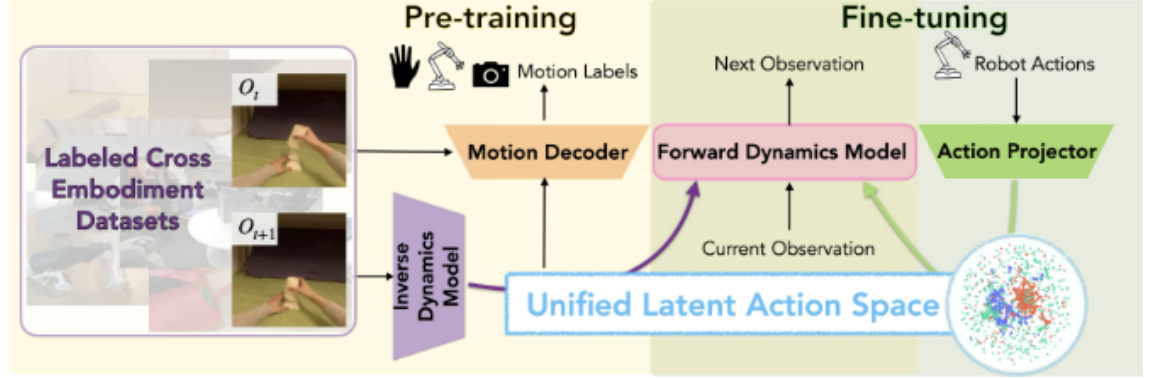


Figure 2.3 LAC-WM (Huang et al., 2026), the architecture this thesis adapts. An inverse model encodes an observation pair into a unified latent action conditioning a forward model, and a motion decoder maps the latent back to each embodiment’s motor labels. The paper’s inverse/forward *dynamics* models are what this project instantiates on visual embeddings and renames Inverse/Forward *Transition* Models (Section 3.4).

Because z_t is supervised partly to help reconstruct e_{t+1} , the ITM could cheat by copying the next frame’s content into z_t instead of learning an abstract action. LAC-WM prevents this with **cross-augmentation**: two independent augmentations of the frame pair are encoded, the ITM sees one while the reconstruction target comes from the other, so a copied pixel-level representation no longer matches the target. This project inherits the mechanism.

LAC-WM reports that its unified latent transfers across embodiments better than an explicit-action baseline, and that performance *scales* with embodiment diversity. Its premise, however, is that the embodiments differ in action format while still decoding into a coordinate that means the same thing on all of them – end-effector pose. Section 2.3 takes up why that premise does not survive the move to legged bodies with different leg counts.

2.1.5 Measuring what a representation contains

Because the claims here are about what a representation carries, the evaluation uses tools that measure exactly that. A **linear probe** trains a simple linear classifier or regressor to predict a property from a representation; keeping it linear means success reflects the quality of the representation rather than the cleverness of the readout, so a probe measures the **presence** of information in an accessible form. Probes are reported against a **chance baseline**, under **grouped cross-validation** so that success cannot come from memorising frames of the same clip, and against a **shuffled-label control** that must collapse to chance.

A probe measures presence but not dominance: a property can be decodable while contributing almost nothing to how the representation is organised. The **silhouette score**

and between-class variance measure that second question. Reporting only one of the two gives the wrong answer, and both are used throughout Chapters 3 and 4. Where the question is not what a representation contains but whether a *model* uses a quantity, presence-style metrics are insufficient on their own – a sharper instrument for exactly that question is what Section 2.5 develops.

2.2 The Cost of Knowing the Body

Making one controller work across bodies is an active problem, and the strategies differ mainly in *what they must be told about the new body before they can control it*. Organised this way rather than by architecture, the literature falls into four groups, and a clear pattern holds across all of them: the body description is always supplied from somewhere.

2.2.1 Retraining, with or without a latent action

The baseline strategy is to train a new controller for each body, carrying nothing across. DreamerV3 (Section 2.1.1) is representative: highly general across *tasks*, but a fresh model per domain. Introducing a latent action does not by itself change this. Li et al. (2021) plan in a *learned latent action space* for legged locomotion and demonstrate it on both a hexapod and a quadruped, but their framework is re-instantiated and trained *separately per robot*: no latent space or dynamics model is shared or transferred between them. It establishes that latent actions suit legged control without making the latent space itself the thing that transfers – the cost of Section 2.1.1 persists even once the action representation is learned rather than native. This is the reference against which every sample-efficiency claim in this thesis is measured.

2.2.2 Explicit morphology conditioning

A second group avoids retraining by being handed a description of the body’s structure. QWM (Danesh et al., 2026) conditions a world model on physical parameters – limb lengths, masses, torque limits – read from each robot’s CAD/USD description, and generalises zero-shot to unseen quadruped morphologies (Figure 2.4). It is the strongest cross-morphology transfer result in this area and the clearest contrast to this thesis: the morphology is *supplied*, not inferred, and its observation channel is proprioception. Notably, QWM routes what it calls “unmodeled real-world residuals” through a separate dynamic latent – an implicit concession that the design file does not fully describe the physical body.

Two related families ask for less than a full design file, but still ask for a structural description. Graph- and transformer-based controllers such as MetaMorph (Gupta et al., 2022) learn morphology-agnostic policies over many procedurally varied bodies by representing the body as a graph of joints, and Ai et al. (2025) study how such generalisation scales with the number of training bodies; these do not need a CAD file, but they must be

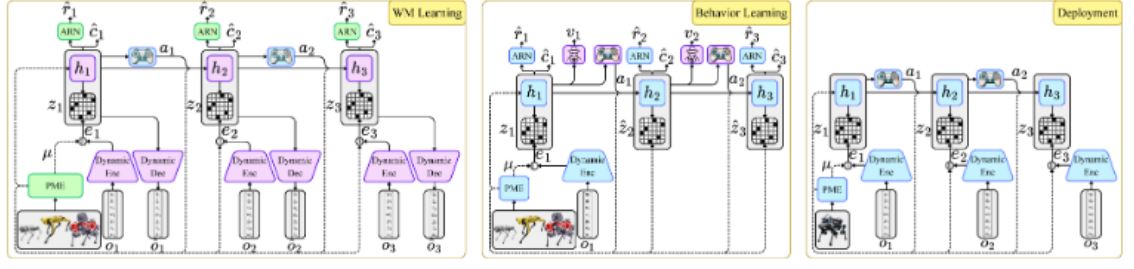


Figure 2.4 QWM (Danesh et al., 2026). A Physical Morphology Encoder derives an embedding from each robot’s USD design file and conditions the world model on it; a policy is then learned in imagination and deployed by injecting that morphology’s embedding. The morphology is supplied from a design file – exactly the privileged input this thesis removes.

handed the **kinematic tree**, which joint connects to which. Online system identification instead infers hidden dynamics parameters from a window of recent interaction, which removes the design file but requires access to the robot’s internal signals and enough new interaction to identify the parameters.

2.2.3 Fitted per-robot adapters

A third group shares a policy *structure* across bodies while still fitting a per-robot translation layer from that body’s own sensing. L3P (Zheng et al., 2025) wraps a shared latent policy backbone in a per-robot observation encoder and per-robot latent-action decoder, so control transfers across diverse legged robots (Figure 2.5). Its decomposition is a close structural parallel to this project’s shared trunk with body-specific heads. The difference is the channel and the direction of information: L3P fits its per-robot adapters from that robot’s own proprioception and foot force, with joint ordering and actuator conventions known.

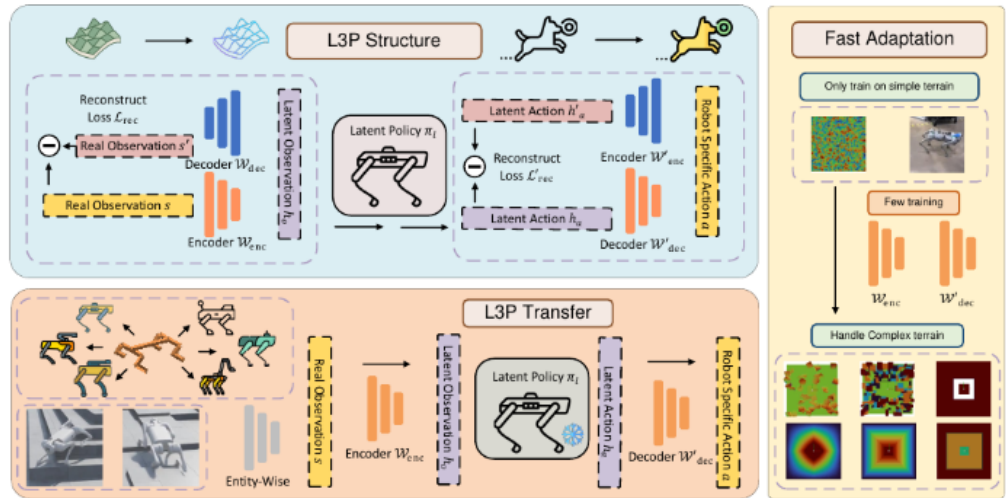


Figure 2.5 L3P (Zheng et al., 2025). A shared latent policy backbone is wrapped by a per-robot observation encoder and a per-robot latent-action decoder. The shared-latent-with-body-specific-heads structure parallels this project’s, but L3P operates on proprioception and must fit an encoder and decoder per robot.

2.2.4 Alignment objectives and paired demonstrations

A fourth group obtains the correspondence from data rather than from the body, and both examples are vision-based. UniSkill (Kim et al., 2025) learns an embodiment-agnostic skill representation from video, including human video, using an inverse/forward skill-dynamics pair (Figure 2.6); its invariance is *optimised for* by training on cross-embodiment data with an alignment objective. Demo-JEPA (He et al., 2026) reframes a demonstration as specifying a future *goal state* rather than actions to copy, and translates source demonstrations into target-compatible latent goals that the target realises through its own world model. It requires neither a shared action space nor action-level retargeting, but the translation is fitted from the target robot’s own interaction experience on trajectories paired with the source – by retargeted end-effector replay in simulation, and by Global Temporal Cycle Consistency alignment on real data.

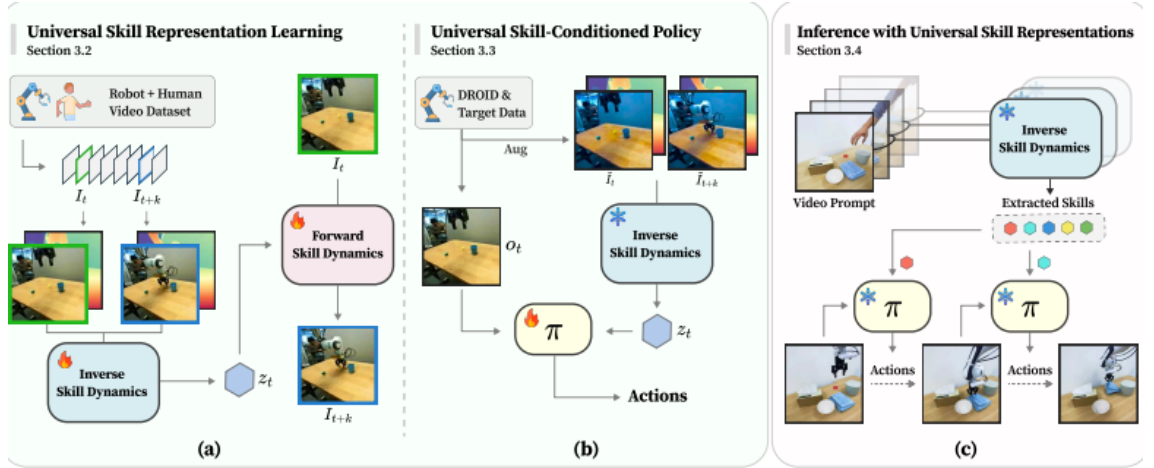


Figure 2.6 UniSkill (Kim et al., 2025). Inverse and Forward Skill Dynamics models are trained on diverse video to encode a body-independent skill representation, which then conditions a policy. The inverse/forward pairing matches this project’s ITM/FTM; the difference is that UniSkill *aligns* its representation across bodies by training for it.

2.2.5 Synthesis: what each method must be told

Table 2.1 places these four groups on the axes that decide this thesis’s claim. Reading down the “morphology information” column, every method obtains the body description from inside the body, from a design file, or from demonstrations paired on the target. Reading down “disjoint action spaces”, the methods that do transfer do so across bodies whose action space is shared by construction, bridged by an adapter fitted with privileged access, or bridged by paired same-task demonstrations. It is also worth noting that the vision-based latent-action methods are demonstrated almost entirely on **manipulation**, where end-effector pose already provides a coordinate common to every body in the corpus. Legged locomotion across different leg counts offers no such coordinate, which Section 2.3 develops.

Table 2.1 What each method must be told about a robot before it can control it. “Disjoint action spaces” asks whether the bodies unified share no dimension or correspondence without external help; “transfers across bodies” asks whether a single trained model is reused on a new body without per-body retraining from scratch.

Method	Observation	Morphology info. from	Action representation	Disjoint action spaces?	Transfers across bodies?	Domain
DreamerV3 (Hafner 2023)	state / pixels	retrained per body	native command	n/a	no (per body)	general control
Li et al. (2021)	proprioception	retrained per body	learned latent action	yes (hex. & quad.)	no (per body)	hexapod, quadruped
QWM (Danesh 2026)	proprioception	CAD / USD file	native command	no (quadruped)	yes, zero-shot	quadruped
MetaMorph (Gupta 2022)	proprioception + graph	kinematic tree	native command	no (procedural)	yes (in family)	procedural bodies
Online system ID	proprioception	interaction	native command	no	yes, adapted	legged / general
L3P (Zheng 2025)	proprioception + force	fitted adapter	shared policy latent	yes (adapter)	partial (adapter)	legged
LAC-WM (Huang 2026)	vision	from transition	learned latent action	yes (arm, hand, human)	yes	manipulation
UniSkill (Kim 2025)	vision	training objective	skill latent	yes (human/robot)	yes	manipulation / human
Demo-JEPA (2026)	vision	paired target demos	latent goal state	not needed (paired)	yes (with target data)	manipulation
Hu et al. (2025)	vision (egocentric)	none needed	image-space model	n/a (one robot)	no (per robot)	legged locomotion
This work	vision (egocentric)	never supplied	shared body-motion coord.	yes (18 vs. 12 DOF)	cheap adapt., not zero-shot	legged locomotion

2.3 What Cross-Embodiment Locomotion Requires

Every method in Section 2.2 resolves the correspondence problem by being handed a mapping from outside: a kinematic tree, a fitted adapter, or paired demonstrations. This section states what a method that is handed nothing must find instead, before Section 2.4 asks how much of the remaining requirement – knowledge of the body’s own structure – can be dropped as well.

2.3.1 A shared coordinate that is not joint space

An 18-joint hexapod and a 12-joint quadruped have action spaces that share no dimension and no correspondence. No relabelling creates one: which of the hexapod’s three joints per leg corresponds to which of the quadruped’s, and what happens to the two extra legs, has no physically correct answer. A method that is handed nothing must instead find a quantity that is *already* the same physical thing on both bodies and use it as the shared interface.

Task-space foot trajectories are the obvious candidate and are rejected here for two independent reasons. Computing them requires a kinematic model and inverse kinematics for every robot, which is the cost this work exists to avoid and which fails immediately on a body whose kinematics are unknown. And empirically, a foot’s motion decomposes into the part that is body motion rewritten – adding nothing not already available – and the gait itself, which is exactly the part that does not correspond across a six-legged wave and a four-legged trot.

What a hexapod and a quadruped genuinely share is at the level of the **body**, not the leg: how fast the body travels forward, how fast it slides sideways, and how fast it rotates. Those three quantities are defined identically on both robots, observable from outside without any kinematic model, and are what this thesis uses as the shared coordinate.

2.3.2 Dynamic similarity and Froude scaling

Comparing those quantities across bodies of different size requires normalising them. The dynamic-similarity hypothesis (Alexander and Jayes, 1983) formalises this for animal gaits: geometrically similar bodies of different scale move equivalently only when compared at matched dimensionless speed, the Froude number $Fr = v^2/(gL)$, where L is a characteristic length. A normalised forward speed $v/\sqrt{gL}$ is therefore the scale-free version of “how fast this body is walking”, and the same reasoning gives dimensionless lateral speed and yaw rate. This is why the shared coordinate in this thesis is Froude-scaled rather than expressed in metres per second: without it, a large quadruped and a small insect walking equivalently would read as different behaviours, and the coordinate would encode body size rather than behaviour.

2.3.3 Our test pair, and why it is a valid dynamic-similarity test

This thesis pairs a simulated stick insect (*Medauroidea extradentata*, source) with a Unitree B1 quadruped (target, MuJoCo). The pair is chosen deliberately, not inherited from prior work: the two share no joint correspondence, no leg count, no gait family, and no simulator, so nothing but the body-level coordinate of Section 2.3.1 can carry information between them – which is what makes the pair a genuine test of that coordinate rather than a convenient one. The insect side is additionally chosen because a validated simulation model of the exact species already exists (Larsen et al., 2023) and its walking is exceptionally well characterised: coordination arises from decentralised local rules between neighbouring legs (Cruse, 1990) rather than a central clock, and at the low, statically-oriented walking speed used here the gait is a phase-staggered wave rather than a clean tripod, consistent with both the recordings and the mature controller of Larsen et al. That level of characterisation is what makes the source body’s behaviour trustworthy as ground truth, not merely convenient to simulate.

2.4 Dropping the Body Description Entirely

Section 2.3 defined what a shared interface between two disjoint bodies must look like. A separate question is whether a controller needs to be told anything about a body’s structure at all, and one result removes that requirement entirely – the closest precedent to this thesis.

Hu, Chen, and Lipson (2025) show that a task-agnostic *visual self-model* – a forward model mapping the current egocentric view and a motor action to the next egocentric view

– can be learned for a legged robot from a single first-person camera and **random motor babbling**, with no prior knowledge of the robot’s morphology, kinematics, or task.

Their argument runs as follows. Legged robots have historically been “blind”, relying on proprioception, with vision reserved for high-level planning; the authors challenge the assumption that vision is too slow and too indirect to model a body’s own dynamics. A self-model learned from vision needs none of the infrastructure its alternatives need – no CAD model or explicit equations, no IMU, no external optical tracker, no morphology prior – which makes it portable in a way that matters when a robot must adapt in place. They then widen that claim in stages: basic locomotion (forward, backward, turning) planned entirely in the visual self-model; robustness to unseen textures and slippery ground; the same method repeated across several quadrupeds and a humanoid; and finally, as the capstone, **autonomous damage detection and recovery** – the robot notices that its predictions no longer match what happens, re-babbles, and retrains its self-model, without being told what broke. Each step widens the same claim: the body description was never necessary.

This thesis adopts that premise directly. Two things separate the two works, and both define what is left to do.

First, the self-model is fitted to one robot at a time. Each body is babbled and modelled on its own; the generalisation the authors report is that the *method* works when repeated on different robots, not that one trained model drives a robot it has never interacted with. There is no shared action representation across bodies, and damage recovery is repair *within* one body’s own model rather than transfer *between* two. Extending the premise across embodiments is precisely what this thesis attempts.

Second, the egocentric viewpoint is adopted but never justified against the alternative. The reasons given are practical – one onboard sensor, portability, and ground texture supplying motion cues – and no comparison against a third-person view is run. For a self-model of a single body that is a reasonable and sufficient basis for the choice. Section 2.5 is why, for a *cross-embodiment* world model, it is not.

2.5 Does a Video World Model Use the Action?

Everything so far establishes *what* a cross-embodiment method must be given – a shared coordinate (Section 2.3) that can, following Hu et al. (Section 2.4), be inferred with no morphology description at all. None of it establishes that a world model conditioned on such a coordinate actually *depends* on it when predicting. That turns out not to be safe to assume, and it is the central methodological question this thesis’s own diagnostic work (Chapter 3) is built to answer, in two parts: first whether the action is used *at all*, and second, once it is, whether it is used *precisely enough* to be worth having. This section develops both halves of that question and shows that neither has been asked in this form before.

2.5.1 Three partial views of the same failure

Three recent results, from three different angles, establish that a video world model can appear action-sensitive by an offline metric while its prediction does not actually depend on the action – without any of them measuring how severe this becomes in a periodic domain such as locomotion, or what causes it (Table 2.2).

AHA-WAM (Cai et al., 2026) observes, in manipulation world-action modelling, that coupling a world model’s prediction to the same frequency as action execution “forc[es] the world branch to model near-term frame variations that are redundant and weakly informative”, and responds by decoupling a low-frequency video planner from a high-frequency action expert. The redundancy is treated as a design constraint to work around rather than a phenomenon to explain.

Yeom et al. (2026) approach it from the representation side, asking what makes video world model latents action-relevant. They find that action information is **legible** in a frozen V-JEPA encoder’s features even though it was never trained with actions ($R^2 \approx 0.40$ frozen, rising to 0.85 with a trained inverse-dynamics head), and note separately that static-environment benchmarks let a model succeed by reading per-frame appearance in place of genuine temporal reasoning.

UWM-JEPA (Radha and Goktas, 2026) identifies the corresponding pathology in the training objective. Because the target frame under teacher-forcing is fixed by the trajectory that actually occurred, a model can predict it well without its prediction depending on the action it was given. The paper states the consequence directly – “action sensitivity itself requires training against counterfactual rather than teacher-forced targets” – and proposes counterfactual targets as the remedy.

Table 2.2 What each action-relevance result establishes, and what this thesis measures on the same question.

Work	What it establishes	What this thesis adds
AHA-WAM (2026)	Adjacent frames in manipulation video are “redundant and weakly informative” for an action-conditioned world model; responds architecturally.	Asks <i>why</i> the redundancy arises in legged locomotion, and whether it is a property of the viewpoint rather than of the frame rate.
Yeom et al. (2026)	Action information is <i>legible</i> in a frozen video encoder not built for the purpose ($R^2 \approx 0.40$ frozen, 0.85 with a head).	Distinguishes legibility from use: a model can make the action more readable in its output while depending on it less.
UWM-JEPA (2026)	Teacher-forced targets let a model satisfy its objective without depending on the action; proposes counterfactual targets.	Examines a domain where the action-independent prediction is already near the achievable ceiling, so the proposed remedy has little remaining gap to close.

2.5.2 The unexamined variable: camera viewpoint

None of the three results above asks the question this thesis is forced to confront: **how severe the redundancy becomes when the observation is periodic, and whether**

the camera viewpoint determines it. Locomotion is periodic by construction, and an external, third-person view of a walking body shows a pose that is itself a function of gait phase – the body’s own posture can stand in for the command that produced it. Whether that makes the action redundant, and whether an egocentric view removes the redundancy, is measured in this thesis (Chapter 3) rather than assumed. It is also the question Hu et al. (Section 2.4) never had to ask, because a self-model of one’s own body has no allocentric alternative to compare against, and no method surveyed in Section 2.2 raises the question at all.

2.5.3 Legibility is not use, and use is not precision

Even where an offline metric shows an inferred action is legible, Section 2.5.1 established that this does not show a world model *uses* it – a distinction this thesis measures directly, showing a model’s output can become *more* legible in exactly the way an offline metric rewards while the model’s actual dependence on the action falls. Establishing that a world model uses the action *at all*, at a single prediction step, is therefore only the first of two questions this thesis’s diagnostic chapter answers, and answering it does not settle the second: whether that dependence is *precise* enough to rank similar candidate actions against each other, or stable enough to remain informative over many steps of rollout rather than only one. Coarse, single-step conditioning and fine-grained, multi-step usability are separate capabilities, and no method reviewed in this chapter distinguishes them – each is evaluated, at most, on whichever single capability its own benchmark happens to test. This chapter’s synthesis (Section 2.6) states the gap this two-part diagnostic question exposes; Chapter 3 reports which of the two capabilities this thesis’s own world model has, and which it does not yet have.

2.6 Research Gap and the Claim of This Thesis

Section 2.2 established that every cross-embodiment method must be told something about the new body: a design file, a kinematic tree, a fitted adapter, or demonstrations paired on the target. Section 2.3 established what a method that is told nothing must find instead – a shared coordinate that is not joint space, Froude-scaled so it encodes behaviour rather than body size. Section 2.4 established that the remaining requirement, knowledge of a body’s own structure, can also be dropped entirely, for a single robot, and that nobody has extended that to a second. Section 2.5 established that even once both of those are granted, a further and largely unexamined question remains: whether the resulting world model genuinely uses the coordinate it is given, at a single step and with enough precision to be useful, and that the influence of camera viewpoint on this is unexamined in the literature.

The gap is the conjunction, and no surveyed method occupies it: **vision-only observation, no kinematic model or design file for either body, genuinely disjoint action spaces (18 against 12 joints, six legs against four), and evaluation through closed-loop**

control rather than an offline number alone.

The claim this thesis makes. A behaviour recorded on one robot can drive a structurally different robot from egocentric video alone – with no URDF, no kinematic model, and no demonstrations of the target body – through a Froude-scaled body-motion coordinate; and the viewpoint from which that world model is built is not a free choice, because in periodic locomotion an external view lets the body’s own pose stand in for the command and the world model never learns to use the action.

What is deliberately not claimed is set out in Section 1.4 and repeated here in one line: not zero-shot transfer, not reliable long-horizon closed-loop control, not a scaling law over many embodiments, and not that a camera is currently the only thing this pipeline needs about a new robot.

Why simulation. Answering this requires ground-truth control over the variables under study: two bodies whose action spaces are verifiably disjoint, driven under known conditions, with exact body pose and contact logged, and the ability to hold the camera and the rendering fixed while only the viewpoint changes. Only simulation provides that, and it is what makes the viewpoint comparison in Chapter 3 a controlled measurement rather than an observation. Real-robot transfer is out of scope (Section 1.4); the contribution is the controlled demonstration of what does and does not cross, and why.

On the observation modality. It is worth stating plainly why vision rather than proprioception, since the answer differs between the two stages of this work. Across three leg-length variants of one hexapod (Stage 1), the three bodies share an 18-dimensional joint space, so proprioception could in principle be shared too and vision is the harder demonstration rather than a necessity. Across the hexapod and the quadruped (Stage 2), there is no proprioceptive coordinate to share without first constructing a joint correspondence from outside – exactly the privileged information this thesis withholds – while one camera describes both bodies in the same pixel array. The strongest version of the argument is not that proprioception is incapable: morphology-agnostic proprioceptive control exists, treating joints as a token set over the kinematic graph. It is that those methods must be handed the graph, and a camera has to be handed nothing. That distinction also decides which bodies the approach could ever reach: animals, robots whose configuration has diverged from their documentation, and hardware acquired without published kinematics are all cases where an external view is not the preferred channel but the only remaining one.

CHAPTER 3 RESEARCH METHODOLOGY

3.1 System Overview

The system comprises four learned/parameterized components on top of a fixed perceptual front-end, trained end-to-end except for the frozen encoder:

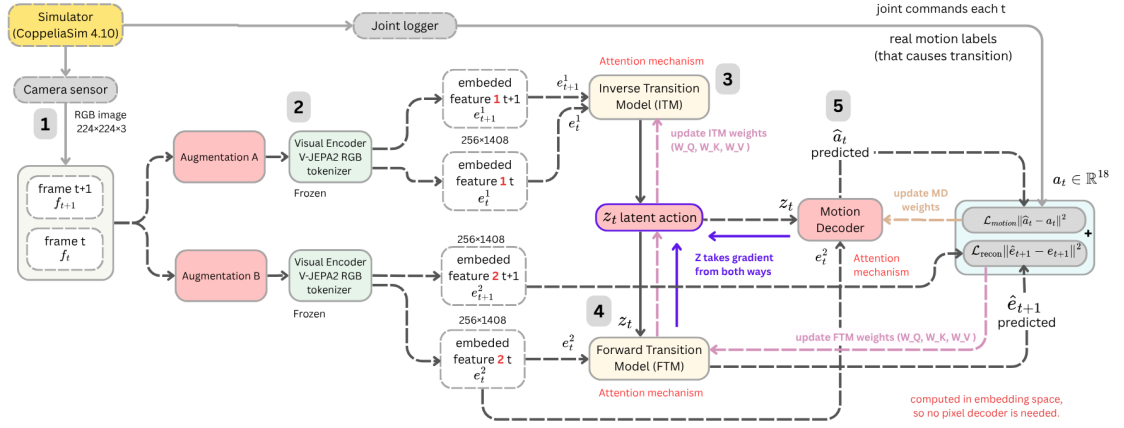


Figure 3.1 System pipeline. A frozen V-JEPA2 encoder maps each RGB frame to an embedding e_t ; the Inverse Transition Model infers the latent action z_t from (e_t, e_{t+1}) ; the Forward Transition Model predicts $\hat{e}_{t+1}$ from (e_t, z_t) (loss $\mathcal{L}_{\text{recon}}$); and the Motion Decoder maps z_t back to a body-specific joint command $\hat{a}_t$ (loss $\mathcal{L}_{\text{motion}}$).

Input/output specification of each block (units stated, per the advisors' requirement):

Table 3.1 Input/output specification of each block in the pipeline.

Block	Input	Output	Trained?
V-JEPA2 encoder	RGB frame, $256 \times 256 \times 3$ (uint8)	$e_t \in \mathbb{R}^{256 \times 1408}$ patch tokens	No (frozen)
Inverse Transition Model	$[e_t, e_{t+1}]$ (512 tokens)	$z_t \in \mathbb{R}^{64}$ (latent action)	Yes
Forward Transition Model	$[e_t, z_t]$	$\hat{e}_{t+1} \in \mathbb{R}^{1408}$	Yes
Motion Decoder	$(e_t$ as context, z_t as query)	$\hat{a}_t \in \mathbb{R}^{18}$ (joint targets, rad)	Yes (weights retained for downstream use)

Losses: $\mathcal{L} = \lambda_{\text{recon}} \mathcal{L}_{\text{recon}} + \lambda_{\text{motion}} \mathcal{L}_{\text{motion}}$, where $\mathcal{L}_{\text{recon}} = \|\hat{e}_{t+1} - e_{t+1}\|^2$ is computed in embedding space (no pixel decoder required) and $\mathcal{L}_{\text{motion}} = \|\hat{a}_t - a_t\|^2$.

Cross-augmentation is applied: two independent augmentations of each frame pair are encoded, one feeding the ITM and the other the FTM target, which prevents the ITM from smuggling raw future-frame content into z_t as a shortcut.

3.2 Simulation Environment and Morphology Variants

3.2.1 Simulator and model

CoppeliaSim v4.10 (Bullet 2.78 engine, 20 Hz timestep). The robot is the *Medau-roidea extradentata* model (six legs, three actuated joints each: ThC/CTr/FTi), adopted from Larsen et al. (2023), with an 18-dimensional joint-position-target action space.

3.2.2 Leg-length variants

Three scenes are generated by scaling the coxa/femur/tibia segments of all six legs by a uniform factor and repositioning downstream joints accordingly: **long** 1.0× (**base**), **medium** 0.75×, **short** 0.5×. Scaling is verified numerically (measured foot-reach ratio matches the target factor for all six legs).

3.2.3 Observation capture

A single fixed vision sensor (256×256 , side view, $\sim 40^\circ$ elevation) is added programmatically so that its pose relative to the robot is *identical across all morphologies by construction*. The floor uses a matte, mildly-textured, non-repeating surface (a controlled choice: both a stark checkerboard and a blank surface were found to corrupt the visual features). The camera tracks the robot’s planar position at fixed height and orientation, keeping apparent size constant.

3.3 Visual Encoder (Frozen V-JEPA2)

The encoder is facebook/vjepa2-vitg-fpc64-256 (ViT-g/16, $\sim 1\text{B}$ parameters), used frozen. Because this checkpoint is natively a 64-frame *video* encoder with 2-frame tubelets, feeding a real clip would let each frame attend to other timesteps and contaminate its embedding. To obtain an independent per-frame embedding, each frame is **duplicated into the minimal 2-frame tubelet** and encoded alone, yielding 256 patch tokens of dimension 1408 per frame. This usage has direct precedent in V-JEPA 2-AC.

3.4 Latent Action Model

The two learned transition models below are instances of the inverse and forward dynamics models described in Section 2.1.3: the ITM infers the latent action from a state transition, and the FTM predicts the next state from the current state and that latent. This project refers to them as **Transition Models** rather than *dynamics* models deliberately. They operate entirely on **transitions between visual embeddings** produced by the frozen encoder. “Transition Model” names precisely what the module computes (a mapping over embedding transitions) and avoids implying that the model estimates the body’s physical dynamics, which it does not. The underlying technique is identical to the inverse/forward dynamics models of the literature; only the name is chosen for descriptive accuracy in the visual-embedding setting.

3.4.1 Inverse Transition Model (ITM)

Attention-based (causal self-attention blocks with a learned query token) mapping the pair $[e_t, e_{t+1}]$ to the latent action $z_t \in \mathbb{R}^{64}$. The causal structure frames z_t as “what happened between t and $t + 1$.” This is the inverse dynamics model of Section 2.1.3, computing z_t from a transition instead of a native action.

3.4.2 Forward Transition Model (FTM)

Attention-based, predicting the next visual embedding $\hat{e}_{t+1}$ from $[e_t, z_t]$; supervised by $\mathcal{L}_{\text{recon}}$ in embedding space.

3.4.3 Motion Decoder (MD)

Cross-attention with z_t as query over the current visual context, producing $\hat{a}_t$; supervised by $\mathcal{L}_{\text{motion}}$ against the auto-logged joint command a_t . This is the only place ground-truth action is used, and the decoder is **not part of the Phase 1 evaluation**, though its weights are retained. Its role during pretraining is to anchor z_t to real motion so the latent cannot collapse into a trivial identity. Beyond this thesis it has a second role: it is the only module that maps a latent action back into executable joint commands, so any downstream policy emitting z_t would require it. This also answers a standing question about the design, namely why a latent is introduced at all when the robot ultimately needs joint commands: the policy operates in the latent space because that is the part expected to transfer across bodies, while the decoder performs the body-specific conversion. The conversion separates what transfers from what does not.

3.4.4 Dimensions and hyperparameters

z_t is 64-dimensional (per LAC-WM §4.2). The λ weights, learning rate, and optimizer are to be determined by ablation (the base paper does not report them). Training uses mixed precision on a single GPU (RTX 2080 Ti, 11 GB), with small batch size and gradient accumulation to fit the frozen 1B-parameter encoder in the training loop (cross-augmentation prevents offline embedding caching).

3.5 Data Collection

3.5.1 Pilot dataset (reference-gait replay)

The preliminary results of Section 3.7 were obtained on a pilot dataset in which each morphology is driven by a single fixed reference gait (replayed real-animal joint trajectories from one *Medauroidea* individual), capturing per timestep the RGB frame, the 18-dimensional joint command a_t , the body pose, and the per-foot contact force. Reloading the scene between episodes provides genuine variation, because the contact-rich dynamics are chaotic and each reload yields a different trajectory, while within an episode the

simulation is deterministic. Behaviour is labelled by the **6-bit foot-contact state** (which of the six feet are planted), a body-pose label preferable to a time-based phase label for the reason given in Section 3.7.3.

3.5.2 Why the pilot dataset is not sufficient for the main experiment

In the pilot, the *same* joint-command sequence is applied to all three bodies (this is exactly what makes the morphology-gap test of Section 3.7.1 valid: identical input, different outcome). Verification confirms the commands are bit-identical across bodies (maximum pairwise difference is zero to machine precision). This property, correct for the gap test, is fatal for training the latent action model. The Motion Decoder is grounded by $\mathcal{L}_{\text{motion}} = \|\hat{a}_t - a_t\|^2$; if a_t is identical across bodies, the target carries no morphology information, the decoder has no reason to condition on the body, and there is nothing to *retarget*. The latent action z_t would then be describable as a body-independent action simply because the *command* was already body-independent, which is a vacuous form of the property this thesis claims to demonstrate. The main experiment therefore requires a dataset in which the *behaviour* is shared across bodies but the *command* differs per body.

3.5.3 Dataset redesign (inverse-kinematics retargeting)

The redesign defines each behaviour not as a joint trajectory but as a **task-space foot trajectory** (a Cartesian path for each foot). Given a shared foot trajectory, inverse kinematics is solved *separately for each morphology* to obtain the joint commands that place that body’s feet on the shared path: $a_t^{\text{body}} = \text{IK}(\text{foot trajectory}, \text{body})$. Because foot position depends on both joint angle and link length, the same foot target yields different joint angles on a long leg and a short leg, so the resulting commands differ per body while the intended behaviour is held constant. This is the precise condition the Motion Decoder must learn to recover: a body-independent intent (the latent action) mapped to body-specific joint commands. The source of the shared foot trajectories is a mature logged expert dataset from Larsen et al. (2023) (`expert_66k_aug3c_fcontact`, 66,000 timesteps) that records foot trajectories and the simulator’s own binary contact labels directly, so the trajectories need not be re-derived and the contact labels do not depend on a hand-chosen force threshold. IK is a closed-form solve requiring no additional training, and it is introduced as an *addition* to the pipeline rather than a change to the model architecture. A single successful IK pipeline also supplies the multiple behaviours (walk, turn, and stop) as additional foot trajectories, which the latent-space validation of Section 3.6.2 requires.

[LIMITATION] *IK produces comparable task-space objectives but does not guarantee identical contact dynamics across morphologies, and the anticipated objection that “learning the Motion Decoder is merely learning IK” is answered by the claim under test, namely that the model recovers this body-specific retargeting from observation alone, without being given the kinematics.*

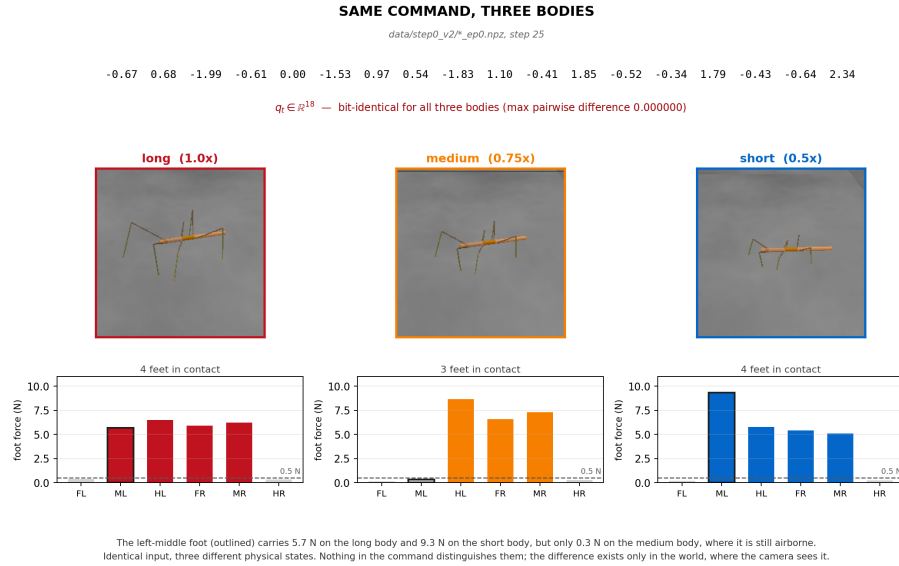


Figure 3.2 A single timestep with a bit-identical joint command across the three bodies. The left-middle foot carries 5.7 N on the long body and 9.3 N on the short body but only 0.3 N on the medium body, where it is still airborne. The command cannot distinguish the bodies; the difference exists only in the resulting physical state.

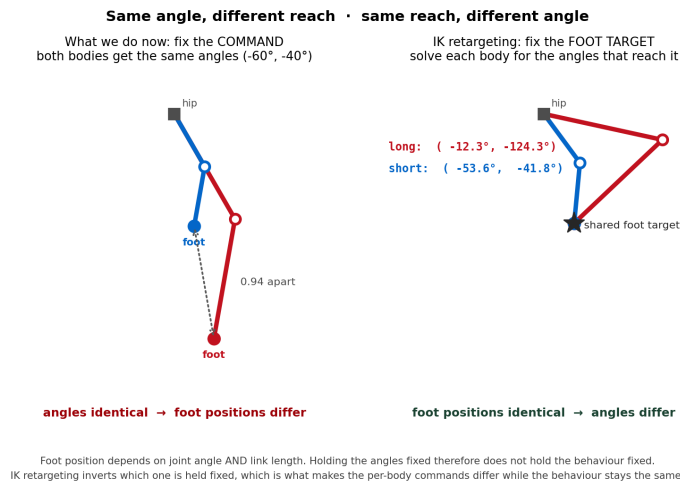


Figure 3.3 Why per-body commands are needed. Left: identical joint angles place the foot in different positions on a long versus a short leg. Right: inverse kinematics inverts the constraint, so a shared foot target is reached by different joint angles per body, which is the body-specific command the Motion Decoder must learn to produce.

3.6 Evaluation Protocol

3.6.1 Metrics

Two complementary measures are reported for every latent-structure claim, since reporting only one can mislead. The first is **linear-probe accuracy**: can a simple classifier decode a label (behaviour or morphology) from the representation? This measures the *presence* of information. The second is the **silhouette score together with between-class variance**, which measure *dominance*, that is, whether that information is the main axis of variation. Cross-morphology **transfer** is measured by training a probe on two morphologies and testing on the held-out one.

3.6.2 Milestone gates

- *Morphology gap*: pass if the three variants travel measurably different distances under identical commands.
- *Encoder sanity*: pass if behaviour (foot-contact state) is decodable from e_t above a noise floor.
- *Latent validation*: pass if z_t lowers morphology decodability and raises cross-morphology behaviour transfer relative to the raw encoder baseline.
- *Transfer*: pass if the pretrained world model reaches the same $\mathcal{L}_{\text{recon}}$ on the unseen morphology with significantly fewer episodes than a from-scratch model.

3.6.3 Primary evidence and the value-over-raw-command check

The **primary evidence** for the thesis is the latent-space validation: does z_t raise cross-morphology behaviour transfer and lower morphology decodability relative to the raw encoder e_t , both at once. This test is clean because it examines z_t on its own: it needs only the held-out body’s video, and it does not compare against the native command.

Does the latent add value over the raw command? This is answered by two measures. The value of the latent is therefore assessed by:

1. **Adaptation efficiency (the primary usefulness metric, per the lab advisor).** Starting from the pretrained world model, adapt to the held-out body and measure the number of training episodes needed to reach a target reconstruction error, against a **from-scratch** model and against a model **pretrained with the raw command** instead of the latent. This measures *how quickly* the representation transfers, which is fair because every model is actually trained/adapted, and it is the metric the lab advisor identified as the main test of whether the world model is useful. The observation-only start (no action) is included to isolate what the action representation contributes.

2. **The availability argument.** The raw command a_t is a *privileged* input: obtaining the correct command for a new body requires that body’s kinematics (through the inverse-kinematics retargeting of Section 3.5.3), which the motivating scenario assumes is unavailable. The latent z_t is inferred from observation and needs no kinematics. A latent that *matches* what the privileged command achieves is therefore already a substantive result, because it recovers from vision alone what otherwise requires the body’s internal description.

A further contrast, where applicable, is an explicit morphology-conditioned model in the spirit of QWM, comparing explicit and implicit morphology handling.

3.7 Preliminary Validation of the Method

The first two milestones of Section 1.5 have been executed on a **pilot dataset** (the reference-gait replay of Section 3.5), and their outcomes justify the design choices above. These are diagnostic results whose purpose is to establish that the methodology is sound before the main training begins; they are **not** the thesis’s final measurements. They will be regenerated on the redesigned IK dataset (Section 3.5), and the exact values are expected to shift. Every figure below is reproducible from the recorded data. Full results belong in the results chapter of the completed thesis.

3.7.1 The morphology gap is measurable and statistically decisive

All three variants were driven with a bit-identical 18-dimensional command sequence, and the body’s world-frame position was read directly from the simulator. Two distance measures are reported: *path length* (the distance actually walked) and *net displacement* (straight-line progress), because the open-loop gait has no heading correction and its side-to-side body oscillation must not be mistaken for a locomotion difference.

Table 3.2 Distance travelled under an identical command sequence, five episodes per morphology.

Morphology	Path length ($n = 5$)	Net displacement ($n = 5$)
Long, 1.0×	5.217 ± 0.069 m	4.404 ± 0.187 m
Medium, 0.75×	4.149 ± 0.019 m	3.569 ± 0.010 m
Short, 0.5×	3.228 ± 0.011 m	2.729 ± 0.011 m

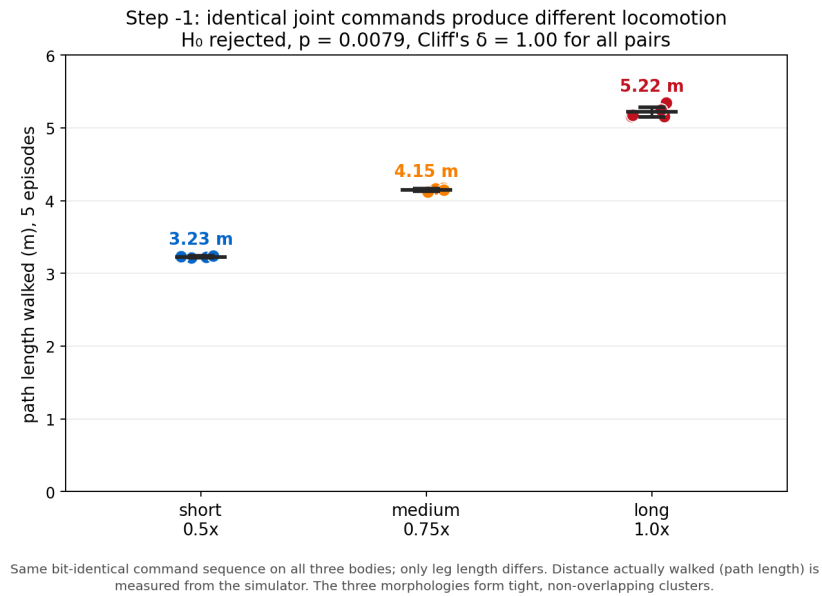


Figure 3.4 Left: top-down body trajectories (five episodes per morphology) under an identical command sequence, aligned to a common start and heading and drawn at true aspect ratio. Right: per-episode path length and net displacement with mean $\pm$ standard deviation; the three morphologies do not overlap.

A two-sided Mann-Whitney U test rejects the null hypothesis of equal outcomes for every pair of morphologies at $p = 0.0079$, with Cliff's $\delta = 1.00$ (complete separation: every episode of a longer-legged body exceeds every episode of the shorter one; the worst long-leg net displacement, 4.032 m, still exceeds the best short-leg value, 2.746 m). At $n = 5$ per group, $p = 0.0079$ is the smallest value the test can return, i.e. the strongest statement this sample size permits. The relationship is not a simple rescaling: walking speed falls with roughly the 0.69 power of leg length rather than the 1.0 that naive geometric scaling would predict. This confirms that identical commands yield genuinely different motion, which is the premise of the whole study.

3.7.2 The frozen encoder carries the required behaviour signal, entangled with body shape

Using a linear probe (logistic regression with standardisation) on the mean-pooled 1408-dimensional embedding, foot-contact state (which feet are planted, top-8 patterns) is decodable **within a body** at a mean macro-F1 of 0.84 (long 0.83, medium 0.95, short 0.78; per-body variance is high, ± 0.18 on the long body, because only five episodes are collected), against a shuffled-label control at chance (macro-F1 ≈ 0.12 , chance = 0.125). The Inverse Transition Model therefore has a usable behaviour signal to extract. The same probe **across bodies** (train on one morphology, test on another) collapses to a mean macro-F1 of 0.16: the behaviour information is present but **entangled with body shape**, and transfer degrades as the leg-length gap widens. Morphology itself is separately decodable from the raw embedding at approximately 100 percent (holding under episode-grouped cross-validation, so it is not frame memorisation), and its leading principal component

orders the three bodies by leg length with no labels supplied. This is the expected and desirable baseline: the encoder plainly sees that the bodies differ, and removing that difference while keeping the behaviour signal is precisely the job of the latent action model. **Two caveats define the pilot status of these numbers:** each morphology was recorded in a single session, so morphology and recording session (lighting, background) are perfectly confounded and cannot be separated on this data; and the top-8 contact patterns cover only 43 percent of frames because the replayed gait is a phase-staggered wave rather than a clean tripod, so the within-body figure is measured on the cleaner, more frequent half of the data and is likely optimistic. The multi-session control needed to attribute the morphology signal to leg length rather than session is scheduled with the dataset redesign.

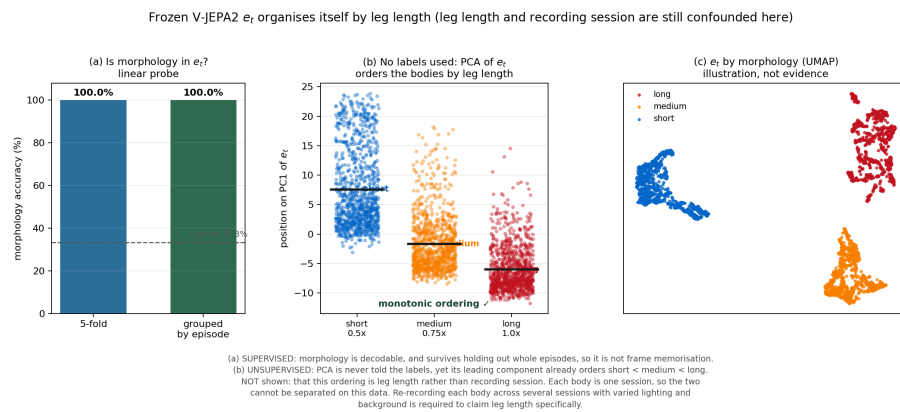


Figure 3.5 Morphology is strongly and ordinally encoded in the frozen embedding. (a) A supervised linear probe reaches ~100 percent and holds under episode-grouped cross-validation. (b) Unsupervised PCA, given no labels, orders the three bodies by leg length along its leading component. (c) A UMAP projection, shown for illustration only. Leg length and recording session are confounded in this pilot.

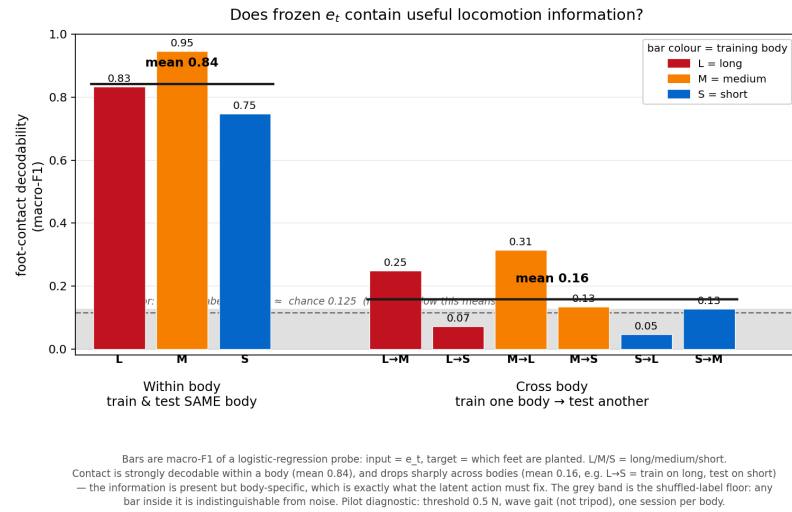


Figure 3.6 Foot-contact behaviour is decodable within a body (mean macro-F1 0.84) but collapses across bodies (mean 0.16), and a shuffled-label control sits at the chance floor. The behaviour information is present but entangled with body shape, the signal the latent action must preserve while making it body-independent.

3.7.3 Two methodological findings that shaped the protocol

First, a representation can *contain* a signal without that signal *dominating* it, so a single metric can mislead. Behaviour is decodable from the raw embedding (probe macro-F1 0.84 within a body), yet morphology is the dominant axis of variation (morphology silhouette +0.0835 against a much weaker behaviour signal, with morphology decodable at ~100 percent). A linear probe measures presence; the silhouette score and between-class variance measure dominance. Section 3.6.1 therefore requires both, and additionally reports balanced accuracy / macro-F1 because the contact-pattern classes are strongly imbalanced. Second, an early time-based behaviour label (step index modulo a hand-chosen gait length) was abandoned: identical commands do not place different bodies in the same pose, so a time index is not a body-independent behaviour label, and it was found to be an artefact of the chosen trim window rather than a physical gait state. A pose-based label (foot-contact state, read from force sensors) is used from this point forward, and Section 3.5 records its residual limitations.

3.8 Remaining Work and Next Steps

The two preliminary milestones of Section 3.7 establish that the morphology gap is real and that the frozen encoder carries a usable, morphology-entangled behaviour signal. This section states, concretely and in order, every step that remains, so that the scope of the committed work is explicit. Each step names what it produces and the condition under which it is considered complete.

Step A: Dataset redesign (inverse-kinematics retargeting). Build the IK pipeline of Section 3.5.3: take the shared task-space foot trajectories from the mature expert log of Larsen et al. (2023) (expert_66k_aug3c_fcontact), solve simIK separately per morphology to obtain per-body joint commands, and drive each body with its own command while recording, per timestep, the RGB frame, the joint command a_t , the body pose, and the simulator’s binary foot-contact labels. *Produces:* a dataset in which behaviour is shared across bodies but the command differs per body. *Gate:* a_t is verified to differ across bodies (variance well above machine precision), removing the pilot’s shared-command limitation of Section 3.5.2.

- *Sub-step A1 (confound control).* Record each morphology across **several sessions** with varied lighting and background, so that a probe trained on one session can be tested on another. This is the control needed to attribute the morphology signal of Section 3.7.2 to leg length rather than to recording session.
- *Sub-step A2 (behaviours).* Generate the walk, turn, and stop trajectories as task-space foot paths (Section 1.4); the latent validation of Step D uses more than one behaviour.

Step B: Regenerate the preliminary analyses on the redesigned dataset. Re-run the morphology-gap and encoder-sanity analyses of Section 3.7 on the IK dataset, using the corrected evaluation of Section 3.6.1 (macro-F1 / balanced accuracy for the imbalanced contact classes, episode-grouped cross-validation, and the multi-session probe from A1). *Produces:* the final baseline values for e_t against which z_t will be compared. *Gate:* the qualitative conclusions of Section 3.7 hold on the clean dataset; the morphology-versus-session attribution is resolved.

Step C: Train the latent action model (ITM + FTM + Motion Decoder). Train the three modules of Section 3.4 jointly with the combined loss and cross-augmentation, on the two training morphologies (short and long). *Produces:* the learned latent action z_t and the trained transition and decoding modules. *Gate:* both losses converge and z_t does not collapse (the Motion Decoder reconstruction stays informative).

Step D: Latent-space validation (the two-sided criterion). Probe z_t for behaviour transfer and for morphology decodability, and cluster z_t by behaviour across bodies. *Gate:* z_t lowers morphology decodability relative to e_t **and** raises cross-morphology behaviour transfer relative to e_t , measured together (Sections 2.6.4, 3.6.1).

Step E: The decisive ablation (run early, not last). Compare a latent-conditioned forward model, $F(e_t, z_t)$, against a raw-joint-conditioned one, $F(e_t, a_t)$, and an observation-only one, $F(e_t, 0)$, under an identical encoder, capacity, dataset, and budget (Sections 2.6.3, 3.6.3). *Gate:* whether the latent improves held-out prediction over the raw command is the central question; this step is scheduled before the transfer experiment so that a negative answer is discovered early.

Step F: Transfer to the unseen morphology. Adapt the pretrained model to the held-out medium body and measure the number of episodes needed to reach a target reconstruction error, against a from-scratch baseline. *Gate:* the pretrained model reaches the target with significantly fewer episodes (the sample-efficiency claim of Section 3.6.3), or the result is reported as negative.

Step G: Analysis, writing, and scoped extensions. Consolidate results. Extensions that the pipeline makes cheap but that are not part of the core claim: an out-of-range (extrapolation) morphology beyond the $0.5\times$ to $1.0\times$ training range, and an explicit cross-embodiment alignment objective in the manner of UniSkill (Section 2.2.4). These are stated as directions, not commitments.

3.9 Work Plan

The timeline follows the steps of Section 3.8. The simulation setup and both preliminary milestones (Section 3.7) are complete; the remaining steps span August to November, with the decisive ablation (Step E) scheduled early so a negative answer would surface before the transfer experiment.

Table 3.3 Work plan. The core experiments (through Step F) target completion by the end of October, leaving November for analysis and writing.

Step	Activity	Aug	Sep	Oct	Nov
(setup)	Simulation setup, morphology variants, vision capture, preliminary milestones (§3.7)	done			
A	Dataset redesign: IK retargeting, multi-session capture (A1), behaviours (A2)	✓			
B	Regenerate the preliminary analyses on the IK dataset (final e_t baselines)	✓	✓		
C	Train the latent action model (ITM + FTM + Motion Decoder)		✓		
E	Decisive ablation: latent vs raw-command vs observation-only (run early)		✓		
D	Latent-space validation (two-sided: behaviour up, morphology down)		✓	✓	
F	Transfer to the held-out medium body; sample-efficiency vs from-scratch			✓	
G	Analysis, thesis writing, scoped extensions			✓	✓

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